

PROBA:

Bayesian Networks

Understanding how one random variable can affect another

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Bullet Lists and Math

We can model coffee orders with **five** variables:

- $t_0 \in \{1, 2, 3\}$
- $r \in \mathbb{R}$
- $p \in \mathbb{Q}$
- $\vec{x} \in \mathbb{R}^2$
- $\hat{y} \in \mathbb{R} \times \mathbb{Q}$

Each variable *is important!*

The shop is doing well, but different baristas excel at different coffees.

When a customer walks in, we want to route the order to the *right* barista.

Tables and Columns

A *Bayesian Network* encodes dependencies as a directed graph, and each node has a CPT.

Root node example:

$P(\text{Weather})$

Weather	P
Cold	0.25
Mild	0.50
Hot	0.25

Non-root example:

$P(\text{Coffee} \mid \text{Tolerance}, \text{Weather})$

(showing 2 of 9 rows)

T,W	Esp	Latte	Flat White
Low, Cold	0.60	0.20	0.20
High, Hot	0.20	0.25	0.55

Boxes and Equations

For a random variable whose domain is $\mathbb{R}$, the following is *always* true:

$$\int_{-\infty}^{\infty} p(x) dx = 1.0$$

Where p is the probability density function of the random variable x .

This is analogous to the the discrete case:

$$\sum_{x \in X} P(x) = 1.0$$

Pseudocode

```
1  define bfs( $G, v$ ):
2      for all  $u \in V$ :
3          | hops[ $u$ ] :=  $\infty$ 
4      hops[ $v$ ] := 0
5      Q.push_tail( $v$ )
6      while Q is not empty:
7          |  $u := Q.pop\_head()$ 
8          | for all edges  $(u, v) \in E$ :
9              | if hops[ $v$ ] =  $\infty$ :
10                 | Q.push_tail( $v$ )
11                 | hops[ $v$ ] := hops[ $u$ ] + 1
```

Diagram from DrawIO

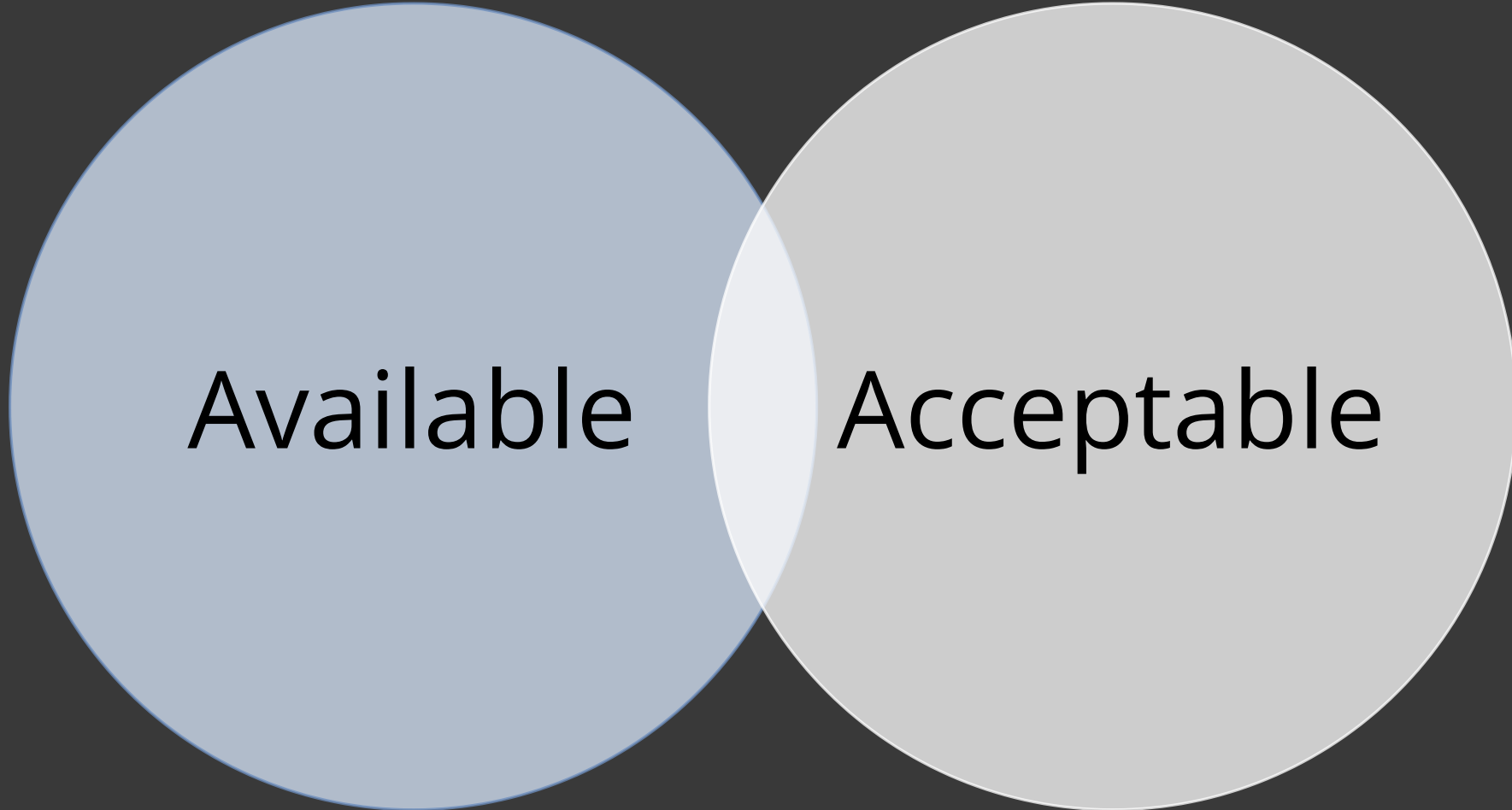




Photo by Basile Morin

“The greatest pleasure of a dog is that you may make a fool of yourself with him, and not only will he not scold you, but he will make a fool of himself too.”

Samuel Butler

Code

```
#include <stdio.h>
```

```
int main() {  
    int a, b, c; // Declare variables  
    a = 3;       // Initialize variables  
    b = 4;  
    c = a + b;    // Do addition  
    printf("%d + %d = %d\n", a, b, c);  
    return 0;  
}
```

Bayesian Networks

Reading Sections 13.1 - 13.3.1

Tomorrow's Lab Making Full Joint Distributions